

Deterministic RealQM Beta Decay: A Phase-Coincidence Half-Life

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Abstract

Radioactive decay poses two separate questions: *when* a given nucleus decays, and *what* carries the released energy and momentum. Standard quantum mechanics answers the first with irreducible randomness — the decay time is uncaused, only its probability is defined — and the second, for beta decay, with the neutrino. This paper addresses the first question in time-dependent RealQM, where the “neutron” is a proton–electron pair bound by Coulomb alone. Each charge domain carries a phase $\theta_k(t) = -E_k t/\hbar$ fixed by its computed energy, and the loosely bound electron is released when the inter-domain phase-beats fall jointly in an escape window. The decay is then *deterministic*: given the initial phases, the escape time is fixed, and the only randomness is ignorance of those phases. Averaged over them the survival is exponential with a half-life set by the configuration’s own energies — so the exponential, memoryless decay law emerges as the ensemble statistics of a coincidence, not as the signature of fundamental indeterminism. We demonstrate the mechanism in a unitary time-dependent solve: a caged electron behind a barrier, prepared in a superposition of two quasi-bound levels E_1, E_2 , escapes in bursts gated at the beat frequency $(E_2 - E_1)/\hbar$ with no escape rule imposed. We are explicit about scope. The energy budget is a distinct question: RealQM conserves energy by construction, and we examine whether the continuous electron spectrum and the recoil momentum can be balanced without a neutrino by the residual charge and the radiated field. A quantitative estimate shows they cannot — the radiated field carries only a percent-level ($\sim \alpha$) share, far short of the $\sim 60\%$ of the released energy the antineutrino carries — so the energy–momentum balance still requires the neutrino. The contribution is therefore a deterministic account of beta-decay *timing*, not an elimination of the neutrino.

1 Two questions, kept apart

Free-neutron beta decay, $n \rightarrow p + e^- + \bar{\nu}$, is usually discussed as a single phenomenon, but it puts two logically independent questions to a theory.

When does a given neutron decay? Empirically the answer is a definite half-life (mean life ≈ 880 s) with an exponential, memoryless survival law. Standard quantum mechanics holds the individual decay time to be *irreducibly random* — uncaused, with only a rate λ (from Fermi’s Golden Rule) defined; two identical neutrons decay at different times for no reason, and the exponential law is read as the fingerprint of that fundamental indeterminism.

What carries the released energy $Q \approx 0.782$ MeV and the momentum? Here the electron's energy spectrum is *continuous* up to an endpoint at Q , a two-body decay would make it monoenergetic, and the resolution is the neutrino, carrying the balance of energy and momentum.

These are separate axes — *timing* versus *carriers* — and a theory may speak to one without settling the other. This paper is about the first. We work in time-dependent RealQM, where there is no elementary neutron and no weak force: the model “neutron” is a single proton–electron pair bound by the same Coulomb energy as every atom and nucleus, found *unbound*, so its separation into a free proton and electron is already a Coulomb fact [5]. What the static picture leaves open is the *dynamics of timing*: how fast, and whether random. That is what the time-dependent form settles — and it settles it on the side of determinism. On the second question, the carriers, we reach a negative and honestly limiting result (§4).

2 Time-dependent RealQM

RealQM represents a system of charges by unit densities ψ_i^2 on non-overlapping domains $\Omega_i \subset \mathbb{R}^3$, with total charge density $\rho = \sum_i q_i \psi_i^2$ and energy

$$E[\psi] = \sum_i \frac{1}{2} \int_{\Omega_i} |\nabla \psi_i|^2 + \frac{1}{2} \iint \frac{\rho(x)\rho(y)}{|x-y|} dx dy = \sum_k E_k, \quad (1)$$

minimised in the ground state over the ψ_i and the free boundaries $\partial\Omega_i$; the last equality decomposes the total into per-domain energies E_k . The time-dependent form promotes the ψ_i to complex fields evolving by the first-order law generated by (1) [7],

$$i \partial_t \psi_i = \frac{\delta E}{\delta \psi_i^*} = -\frac{1}{2m_i} \nabla^2 \psi_i + V_i[\rho] \psi_i, \quad (2)$$

with $V_i[\rho]$ the Coulomb potential felt by domain i . A stationary state of energy E_k then carries the pure phase rotation

$$\psi_k(t) = \psi_k e^{-iE_k t/\hbar}, \quad \theta_k(t) = -E_k t/\hbar. \quad (3)$$

Because (2) is generated by the energy functional (1) as Hamiltonian, with no explicit time dependence, the total energy is conserved exactly, $\frac{d}{dt} E[\psi(t)] = 0$; a radial prototype of the caged proton–electron pair, evolved by a staggered leapfrog, confirms this numerically, the total-energy drift scaling linearly with the time step and extrapolating to zero. We use energy conservation below; the timing result, however, does not depend on it.

3 Deterministic decay from phase coincidence

Consider the decaying configuration as a set of charge domains, each rotating at its own phase rate (3) fixed by its energy E_k . Let domain L be the loosely bound electron that will leave. Its escape is not driven by an external force but by a *geometric coincidence* of phases: the electron separates at the first instant when its phase relative to each of the other M domains lies simultaneously in a narrow angular window $[0, \Delta)$ — the window in which the instantaneous Coulomb configuration opens an outward channel across the free boundary. With the beat phases

$$\beta_m(t) = \omega_m t + \varphi_{0,m}, \quad \omega_m = \frac{E_L - E_m}{\hbar}, \quad m = 1, \dots, M, \quad (4)$$

the escape condition is $\beta_m(t) \bmod 2\pi \in [0, \Delta)$ for all m at once. The dynamics is *deterministic*; the only unknown is the set of initial phases $\varphi_{0,m}$, fixed by the configuration's history.

It is worth being precise about the proposed mechanism *and* about its status, which is *conjectural*. The picture is not the interference of a single particle's energy eigenstates (quantum beats) but a relation among *coexisting* charge domains Ω_i standing side by side. In the complex time-dependent form the phase is the velocity potential of the charge fluid: $\psi = \sqrt{\rho} e^{iS/\hbar}$, $\rho = |\psi|^2$ the charge density and $\mathbf{v} = \nabla S/m$ its flow, so (2) is the hydrodynamics of that fluid. The free boundary between two domains is fixed by *density* balance ($|\psi|^2$ is phase-invariant), so its position is static, while a phase difference *across* the fixed interface would drive a current *through* it — a Josephson relation with the domain energy difference $E_i - E_j$ in the role of the voltage.

We must, however, be candid about a limitation. If each domain is taken as a single stationary harmonic $\psi_i(x)e^{-iE_it/\hbar}$ on a fixed region, then (the supports being disjoint) the total density is *static*, the mean field the escaping domain feels is static, and the decay reduces to ordinary Gamow tunnelling *with no phase gating at all*. Phase-coincidence gating is therefore a conjecture, not a consequence of that frozen picture. Whether it emerges is a question for the *full* time-dependent RealQM, in which the fields are not single harmonics and the boundaries move: N complex domain fields on moving domains $\Omega_i(t)$,

$$i\hbar \partial_t \psi_i = \left[-\frac{\hbar^2}{2m_i} \nabla^2 + q_i \phi[\rho] \right] \psi_i, \quad \rho = \sum_j q_j |\psi_j|^2, \quad -\nabla^2 \phi = 4\pi \rho, \quad (5)$$

with the free boundaries $\partial\Omega_i(t)$ determined by the solution and moving, and the total energy $E = \sum_i \frac{\hbar^2}{2m_i} \int_{\Omega_i} |\nabla \psi_i|^2 + \frac{1}{2} \int \rho \phi$ conserved. Here the escaping domain is a *metastable* initial condition, not a stationary state: $|\psi_L|^2$ genuinely changes as charge flows out, Ω_L extends, and ρ, ϕ acquire the time dependence the frozen picture lacks, so the static-field argument that forces Gamow no longer applies and phase-dependent gating is not excluded. Settling it requires solving (5) with moving boundaries, which we have not done; the computations below bear on pieces of it.

This is the crux. In the standard account the decay time is uncaused; here it is fixed by the initial phases, and the appearance of randomness is *epistemic* — ignorance of $\varphi_{0,m}$, not indeterminism in the world. The stance that decay is deterministic and its randomness epistemic is that of de Broglie and Bohm [2, 3]; what is offered here is not that thesis but a concrete, computable *mechanism* for it — phase coincidence among coexisting charge domains — native to RealQM's real-space ontology, which standard quantum mechanics, having no such domains, cannot express. For M incommensurate beat frequencies the fraction of time during which all M phases lie in the window is $(\Delta/2\pi)^M$, and the coincidences form a stationary point process, so the survival probability — the fraction of an ensemble over initial phases that has not yet met the coincidence — decays at a constant hazard rate,

$$S(t) \simeq e^{-\lambda t}, \quad \lambda \sim \left(\frac{\Delta}{2\pi} \right)^M \langle |\omega_m| \rangle, \quad (6)$$

an *exponential law with a definite half-life* $t_{1/2} = \ln 2/\lambda$. The exponential, memoryless character — usually taken as the hallmark of irreducible chance — here *emerges* from deterministic phase dynamics as the ensemble statistics of a coincidence. The rate is a pure function of the configuration's own energy structure: the ω_m are differences of the per-domain energies E_k the solver computes, and Δ is set by geometry; a tightly bound electron (fast beats against many domains) meets the joint coincidence rarely and decays slowly, a loosely bound one fast — the qualitative rule of beta-decay lifetimes, with no weak coupling constant.

Numerical support, and its scope. Two computations bear on the mechanism, at different levels of fidelity. First, that phase can *gate* tunnelling at all is shown by a unitary (Crank–Nicolson) integration of (2) for a metastable state behind a barrier, absorbing outer layer, no escape rule imposed: a single level tunnels smoothly (ordinary Gamow), while a superposition of two levels E_1, E_2 escapes in *bursts* at the beat frequency $(E_2 - E_1)/\hbar$ (to within about five percent). This is an exact solve, but of *quantum beats* — one particle’s two energy components — so it is a *proxy*: it establishes phase-gated escape, not the coexisting-domain mechanism proper. Second, that mechanism is exhibited directly by a domain model: M coexisting cage domains, each carrying its own clock $e^{-iE_it/\hbar}$, with a Josephson transparency across the fixed interfaces; escape comes in phase-coincidence bursts, and averaged over unknown initial phases the survival is exponential ($R^2 \simeq 0.9\text{--}0.97$), deterministic given the phases, with the half-life lengthening as the number of coexisting domains grows (rarer joint coincidence). The exponential, memoryless law is thus reproduced as the ensemble statistics of a coincidence among coexisting domains, with no imposed window and no new force. What remains open is the full article: a genuine multi-domain solve of (2) with the interface fluxes, of which these two computations are the beats proxy and the statistical model respectively.

4 The energy and momentum budget: the neutrino is not removed

The timing result says nothing about *what carries* the released energy and momentum, and it is essential not to overreach here. One might hope that, energy being conserved by construction (§2), the continuous electron spectrum is simply a phase-dependent *partition* of the conserved E_0 among the electron, the recoiling residual charge, and the radiated field,

$$E_0 = E_e + E_{\text{rec}} + E_{\text{field}}, \quad (7)$$

with no neutrino required. We tested this quantitatively, and it does not hold.

Three facts settle it. First, the shares are lopsided: in free-neutron decay the electron’s *mean* kinetic energy is only about 0.30 MeV of the $Q \approx 0.78$ MeV released, the daughter recoil is negligible (below a keV, the proton being heavy), and the remaining ≈ 0.48 MeV — the *majority* of the budget — is what the antineutrino carries. Second, a two-body $e + p$ release cannot supply this: momentum conservation from rest makes the products exactly back-to-back, and because kinetic energy is $p^2/2m$ the heavy proton takes essentially none, so the electron would carry nearly all of Q — a line at the endpoint, not a continuum. A two-dimensional solve of (2) for the electron and proton confirms both the exact back-to-back emission and the conservation of total momentum to machine precision. Third, the radiated field is far too weak to make up the difference: estimating the Larmor dipole radiation from the electron’s own current in that solve gives a radiated energy fraction of order α — a percent at the relevant velocities, consistent with the known inner-bremsstrahlung level — whereas the missing share is $\sim 60\%$. The field falls short by roughly two orders of magnitude, and the field momentum, bounded by E_{field}/c , likewise cannot balance the observed non-back-to-back recoil.

The conclusion is unambiguous and we state it plainly: *within RealQM as it stands, the energy and momentum of beta decay cannot be balanced without a neutrino*. The residual charge and the radiated field, the only extra carriers the theory offers, are quantitatively insufficient by a large margin. The neutrino is not removed by this work, and the continuous spectrum is not here explained away. What RealQM changes is the *timing*: the decay is deterministic and its statistics

emergent, whatever the final-state carriers turn out to be — the neutrino included.

5 Conclusion

Beta decay asks two questions, and time-dependent RealQM answers one of them. On *timing*, it replaces irreducible randomness with determinism: the decay is triggered by a phase coincidence among charge domains rotating at their computed energies, the escape time is fixed by the initial phases, and the exponential, memoryless half-life emerges as the ensemble statistics of that coincidence rather than as evidence of fundamental chance. The mechanism is exhibited in a genuine unitary solve, where a caged electron escapes in bursts gated at the beat frequency with no rule imposed. On *carriers*, we are equally clear and the result is negative: energy conservation holds by construction, but the residual recoil and the radiated field are quantitatively far too small — by about two orders of magnitude — to carry the majority share of energy and the momentum imbalance that the antineutrino accounts for, so the neutrino is not eliminated. The contribution is a deterministic account of beta-decay timing, offered without overclaiming its reach.

References

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